

Research Report

2026 S.T. Yau High School Science Award (Asia)

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Title of Research Report

Hybrid Neural Force Correction for the Unrestricted Three-Body Problem

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Abstract

We present **SIMON**, a hybrid integrator combining a second-order leapfrog scheme with a learned scalar pairwise force correction and two physical constraints: an analytic force direction that preserves Newtonian radial symmetry, and adaptive sub-stepping that resolves close gravitational encounters. Through a controlled four-run ablation study—removing each component in isolation—we show that long-horizon orbital stability over 100 years depends on these structural constraints: adaptive sub-stepping eliminates the timestep-frontier ejection failure ($105\times$ worst-case frontier error reduction), while the learned-direction baseline demonstrates, for the tested initial condition, that replacing analytic direction with a neural approximation produces unbounded trajectories. The neural network correction itself provides an 8.5% accuracy improvement without contributing to stability. Across seven initial conditions spanning near-circular to tight-binary dynamical regimes, **SIMON** maintains bounded dynamics for all stable configurations (mean divergence rate $\lambda = 0.092 \pm 0.068 \text{ yr}^{-1}$) and correctly identifies physical instabilities in all ejecting configurations; 96% bounded/ejected agreement across 500 randomly generated configurations confirms physically trustworthy characterisation beyond the hand-selected regimes. Energy drift is attributable to the leapfrog integration scheme rather than the neural correction; an energy-drift sweep defines the feasible operating timestep range for a given configuration, and the speed–accuracy frontier then identifies the optimal operating point within that range. Applied to the tested configurations, this methodology yields $1.1\times$ to $2.2\times$ speedup over **ias15**. Validation on the real Sun–Earth–Moon system using NASA/JPL Horizons initial conditions confirms that bounded dynamics and the essential role of adaptive sub-stepping extend to real astronomical data. These results demonstrate that long-horizon stability in neural gravitational integrators requires two structurally enforced physical constraints; an integrator built on this principle reliably characterises three-body dynamics and systematically finds the optimal operating parameters for any given configuration class.

Keywords: Three-Body Problem; Neural Networks; Leapfrog Integration; Adaptive Sub-Stepping; Chaotic Dynamics; Gravitational N-Body Simulation; Lyapunov Divergence

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Table of Contents

Abstract	i
Acknowledgement	ii
Commitments on Academic Honesty and Integrity	iii
1 Introduction	1
2 Literature Review	2
2.1 From Two-Body Solvability to Three-Body Non-Integrability	3
2.2 Chaos and Long-Horizon Prediction	3
2.3 Restricted and Unrestricted Three-Body Problems	4
3 Computational Modelling	4
3.1 Learned Simulators and Research Gap	5
4 SIMON: Architecture and Methodology	6
4.1 Method Overview and Reference Solver	6
4.2 Scalar Force-Correction Model	7
4.3 Hybrid Integration and Adaptive Sub-Stepping	8
4.4 Training Procedure	9
4.5 Evaluation Strategy and Scope	10
5 Results	10
5.1 Results I — Stability Decomposition	11
5.1.1 The Four-Run Ablation Design	11
5.1.2 Run A: Adaptive Sub-Stepping Is the Primary Stability Mechanism	11
5.1.3 Run B: Analytic Force Direction as a Structural Physical Constraint	13
5.1.4 Run C: The Neural Network Correction Is an Accuracy Mechanism	15
5.1.5 Synthesis: The Two-Constraint Principle	16
5.2 Results II — Generalisation Across Initial Conditions	17
5.2.1 Seven Initial Conditions Spanning the Dynamical Spectrum	17
5.2.2 Bounded Configurations: Four ICs	18
5.2.3 Physically Unstable Configurations	19
5.2.4 Population-Scale Generalisation Validation	20
5.3 Results III — Physical Consistency and Reproducibility	20
5.3.1 Energy Conservation	20
5.3.2 Timestep Sensitivity of Energy Drift	22
5.3.3 Training Seed Robustness	23

5.4	Results IV — Computational Performance	24
5.4.1	Speed–Accuracy Frontier on IC1	24
6	Discussion and Conclusion	26
6.1	The Two-Constraint Principle	26
6.2	Positioning SIMON Relative to ias15	27
6.3	Limitations and Future Work	28
Appendix A: Physical Validation: Sun–Earth–Moon		30
Appendix B: Force-Level NN Validation		32
Appendix C: SIMON Architecture Schematic		33

1 Introduction

The Three-Body Problem—the gravitational dynamics of three mutually interacting massive bodies—is a classical problem in physics because its trajectories are generally chaotic and highly sensitive to initial conditions; long-horizon prediction must therefore be performed numerically. This remains important in astrophysical applications such as exoplanet stability, stellar cluster dynamics, and spacecraft trajectory design [Gardner et al. \[2006\]](#).

High-accuracy numerical integrators such as `ias15` provide reliable reference solutions, but their step-by-step structure makes long simulations computationally expensive [Rein and Spiegel \[2014\]](#). Neural or hybrid methods offer a possible route to faster simulation [Ulibarrena et al. \[2024\]](#), but unconstrained learned force models can lose physical stability during close encounters, where small position or force errors are strongly amplified [Dehnen \[2001\]](#).

This study addresses the following research question:

What structural constraints must a neural force-correction integrator enforce to achieve long-horizon orbital stability in a chaotic three-body system, and how can such an integrator be used to find the maximum achievable speedup over traditional solvers for a given dynamical configuration?

To investigate this question, we developed **SIMON**, a hybrid integrator combining a second-order leapfrog scheme with a learned scalar pairwise force correction, analytically enforced Newtonian force direction, and adaptive sub-stepping during close encounters.

The key contributions of this work are:

1. **A two-constraint principle for neural gravitational integration:** long-horizon stability requires two structurally enforced physical constraints—analytic Newtonian force direction and adaptive close-encounter timestep resolution. The neural correction contributes accuracy but not stability; neither constraint can be replaced by the learned component. This principle applies to any neural force-correction integrator in a chaotic gravitational system.
2. **SIMON as a reliable dynamical characterisation framework:** built on this principle, **SIMON** serves as a usable tool for characterising three-body configurations: a researcher obtains bounded/ejected classification, divergence rates, and trajectory behaviour across a spectrum of dynamical regimes. Evaluated across seven distinct regimes and 500 randomly generated configurations, **SIMON** achieves 96% bounded/ejected agreement with `ias15`, confirming that these outputs are physically trustworthy across the broader configuration spectrum.

3. **A methodology for finding the optimal operating parameters:** for any bounded configuration, an energy-drift sweep across timestep values identifies the range of dt for which energy conservation remains physically valid; within that range, the speed–accuracy frontier reveals the trade-off between computational speedup and trajectory accuracy relative to `ias15`, allowing the user to select the optimal operating point. Applied to the tested configurations, this yields $1.1\times$ to $2.2\times$ speedup over `ias15`.
4. **Physical consistency of the neural correction:** energy drift across all tested configurations is attributable to the leapfrog backbone rather than the neural correction, with timestep-dependent scaling confirmed independently of NN invocation. This isolates the neural component as an accuracy mechanism and confirms that it introduces no additional energy error beyond the underlying integration scheme.

The paper is structured as follows. Section 2 reviews the physics background of three-body dynamics, chaos, and long-horizon prediction. Section 3 discusses computational modelling, learned simulators, and the research gap addressed by `SIMON`. Section 4 defines the `SIMON` architecture, training procedure, evaluation strategy, and scope. Section 5 presents experimental results across four themes: stability decomposition, generalisation, physical consistency, and computational performance. Section 6 discusses implications, limitations, and future directions. Appendix A provides a real Sun–Earth–Moon validation using NASA/JPL Horizons state vectors, and Appendix B isolates the learned scalar correction through a force-level accuracy test.

2 Literature Review

The literature relevant to this project lies at the intersection of celestial mechanics, numerical integration, and scientific machine learning. Classical mechanics explains why the unrestricted three-body problem is sensitive to initial conditions and difficult to solve analytically, while numerical-analysis literature shows that long-horizon behaviour depends strongly on whether an algorithm preserves geometric structure and handles close encounters robustly Hairer et al. [2006], Rein and Tamayo [2015], Rein and Spiegel [2014], Boekholt and Portegies Zwart [2015]. More recent machine-learning work shows that neural networks can act as learned simulators or learned Hamiltonian surrogates, but also that their success depends on the physical inductive biases built into the architecture Chen et al. [2018], Battaglia et al. [2016], Greydanus et al. [2019], Gruver et al. [2022], Horn et al. [2025]. This section gives the physics background needed to motivate the computational gap addressed by `SIMON`.

2.1 From Two-Body Solvability to Three-Body Non-Integrability

The Two-Body Problem is analytically tractable because the mutual gravitational attraction between two isolated point masses can be reduced to an equivalent one-body problem in a central potential. Newton’s law of universal gravitation gives the magnitude of the force as

$$F = G \frac{m_1 m_2}{r^2}, \quad (1)$$

where G is the gravitational constant, m_1 and m_2 are the two masses, and r is their separation. This framework explains the classical Keplerian orbits used to model systems such as planets, moons, and satellites [Murray and Dermott \[1999\]](#).

The unrestricted Three-Body Problem is fundamentally different. Each body is accelerated by the other two at the same time, so the equations of motion are coupled and nonlinear. For bodies with positions $\mathbf{r}_i$ and masses m_i , the Newtonian equation of motion can be written compactly as

$$\ddot{\mathbf{r}}_i = G \sum_{j \neq i} m_j \frac{\mathbf{r}_j - \mathbf{r}_i}{|\mathbf{r}_j - \mathbf{r}_i|^3}, \quad i = 1, 2, 3. \quad (2)$$

Unlike the two-body case, this system has no general closed-form solution. Its behaviour depends sensitively on mass ratios, separations, and velocities, making numerical simulation the standard approach for studying generic three-body dynamics [Murray and Dermott \[1999\]](#).

2.2 Chaos and Long-Horizon Prediction

A central difficulty of the Three-Body Problem is chaos. Small differences in initial position or velocity can grow rapidly, causing two initially similar trajectories to diverge over long times. One common way to describe this behaviour is the Lyapunov exponent λ , where an initial separation $\delta(0)$ grows approximately as

$$\delta(t) \approx \delta(0) e^{\lambda t}. \quad (3)$$

A positive value of λ indicates exponential sensitivity to initial conditions [Ott \[2002\]](#). This idea traces back to Poincaré’s analysis of the three-body problem, which showed that gravitational systems can exhibit intricate, non-repeating trajectories rather than simple closed-form motion [Poincaré \[1890\]](#).

For this project, chaos has an important methodological consequence. In a chaotic system, long-term pointwise trajectory agreement is not always the right standard for judging a simulator, because even small numerical differences can produce phase drift. Instead, a useful evaluation must also consider qualitative dynamical behaviour: whether a system

remains bounded, whether ejection occurs, whether energy drift is physically explainable, and whether divergence grows at a controlled rate. This motivates the use of divergence-rate proxies and boundedness checks later in this paper.

2.3 Restricted and Unrestricted Three-Body Problems

The Restricted Three-Body Problem is a simplified version in which two massive bodies influence a third body whose mass is small enough to neglect. This formulation led to important results such as Euler’s and Lagrange’s special solutions and the five Lagrange points [Bistafa \[2021\]](#), [Lagrange \[1772\]](#). These ideas are important in practical mission design; for example, the James Webb Space Telescope operates near the Sun–Earth L_2 region [Gardner et al. \[2006\]](#).

However, the present work focuses on the unrestricted problem, where all three bodies exert mutual gravitational forces. This case is more directly relevant to testing neural-assisted integrators because the system does not reduce to a fixed background potential and can include strong pairwise encounters. The unrestricted setting therefore provides a stricter test of whether a learned force model can remain dynamically usable when all interactions are active.

The key implication for this project is that unrestricted three-body dynamics combine three features that make long-horizon simulation difficult: nonlinear mutual forcing, sensitivity to initial conditions, and close encounters where the Newtonian force changes rapidly with separation. These features make purely pointwise long-term trajectory matching unreliable and motivate the computational question addressed next: how can a hybrid neural integrator reduce computational cost while preserving the physical structure needed for stable qualitative dynamics?

3 Computational Modelling

Since unrestricted three-body dynamics do not have a general closed-form solution, their evolution must be computed numerically. A traditional integrator advances the system step by step, so long simulations require many force evaluations, and close encounters can amplify small numerical errors into large trajectory differences. High-accuracy tools such as `REBOUND` and `ias15` are therefore essential benchmarks for this project, because they show what reliable long-horizon gravitational integration should look like [Rein and Liu \[2012\]](#), [Rein and Spiegel \[2014\]](#). At the same time, numerical-analysis literature shows that long-time reliability is not determined only by local truncation error: preserving geometric structure and resolving close interactions are central to physical reliability [Hairer et al. \[2006\]](#), [Boekholt and Portegies Zwart \[2015\]](#).

Neural networks offer a possible route to reducing this computational burden, but the role of the network must be defined carefully. Some learned-surrogate approaches train a model to map from an initial state directly to a future state, thereby skipping some or all intermediate integration steps. **SIMON** does not use this direct state-to-future-state strategy. It remains a timestep-based hybrid integrator, using a neural network only as a local scalar force-correction mechanism during close encounters. The aim is therefore not to replace numerical integration entirely, but to test whether a learned component can reduce computational cost while preserving the physical structure needed for stable long-horizon rollouts.

This distinction is important because close encounters are the most fragile part of the problem. Under Newtonian gravity, the force between two bodies separated by distance r is proportional to $1/r^2$, so the force magnitude and its gradient change rapidly when r becomes small. A small learned error in position, force magnitude, or force direction can then produce a large artificial impulse, causing non-physical energy change or ejection-like behaviour [Dehnen \[2001\]](#). The central modelling question is therefore not simply whether a neural network can approximate a three-body trajectory, but which parts of the Newtonian force law must remain exact for a learned integrator to remain dynamically usable.

3.1 Learned Simulators and Research Gap

Prior work gives three important lessons for this project. First, classical integrators provide the reliability standard: **WHFast** demonstrates the value of fast symplectic long-term orbit integration, while **ias15** provides a high-order adaptive Gauß–Radau reference method for gravitational dynamics [Rein and Tamayo \[2015\]](#), [Rein and Spiegel \[2014\]](#). Work on the reliability of N -body simulations further shows that chaotic three-body scattering can require very high numerical precision to obtain converged individual trajectories [Boekholt and Portegies Zwart \[2015\]](#). This makes close encounters and long-horizon agreement especially demanding tests for any hybrid method.

Second, learned simulators show that neural networks can approximate dynamical systems, but they do not by themselves guarantee physical reliability. Neural ODEs learn a continuous-time derivative field [Chen et al. \[2018\]](#), while Interaction Networks and graph-network simulators learn object interactions and roll out multi-body systems over time [Battaglia et al. \[2016\]](#), [Sanchez-Gonzalez et al. \[2020\]](#). In celestial mechanics, Breen *et al.* showed that deep networks trained on high-precision three-body solutions can approximate bounded-time three-body evolution at fixed cost [Breen et al. \[2020\]](#). Physics-informed neural networks take another route by imposing governing equations through the loss function [Cuomo et al. \[2022\]](#), including recent applications to the three-body problem [Pereira et al. \[2025\]](#). These works show the promise of learned dynamics, but

they do not isolate which components of the Newtonian force law should be learned and which should remain analytically enforced.

Third, structure-preserving machine learning shows that adding physical inductive bias can improve extrapolation. Hamiltonian and Lagrangian Neural Networks encode mechanical structure through energy-based formulations [Greydanus et al. \[2019\]](#), [Cranmer et al. \[2020\]](#), while symplectic or Hamiltonian-inspired architectures aim to preserve geometric properties of the underlying dynamics [Zhong et al. \[2020\]](#), [Jin et al. \[2020\]](#), [Tong et al. \[2021\]](#), [David and Méhats \[2023\]](#), [Horn et al. \[2025\]](#). However, later analysis also shows that not every physics-inspired bias contributes equally; some apparent Hamiltonian-network advantages may arise from other inductive biases such as second-order structure [Gruver et al. \[2022\]](#). This supports a more targeted question: in a chaotic gravitational system, which physical constraints are actually necessary for stable long-horizon rollouts?

This question is especially important because exact pointwise agreement over long horizons is often unrealistic in chaotic systems. A useful learned model should remain in the correct qualitative dynamical regime, rather than only minimising short-horizon rollout error [Linot et al. \[2022\]](#), [Margazoglou and Magri \[2023\]](#). For the three-body problem, this means preserving bounded motion when the reference solution remains bounded, identifying ejection when the reference system becomes unstable, and keeping energy drift physically interpretable.

The gap addressed by **SIMON** is therefore specific: prior work shows that learned simulators can be fast and that physical inductive biases can improve generalisation, but it does not clearly test whether a neural force model should learn the full gravitational force vector or whether Newtonian radial direction should remain analytic while learning is restricted to a lower-dimensional scalar correction. Nor does it usually separate the stabilising effect of close-encounter timestep refinement from the effect of the neural model itself. **SIMON** is designed around this gap. It combines a leapfrog backbone with analytically enforced radial force direction, adaptive sub-stepping during close encounters, and a neural network limited to a scalar pairwise force correction. Its novelty is methodological: through controlled ablation, it tests which physical constraints are necessary for dynamically usable long-horizon three-body rollouts.

4 **SIMON: Architecture and Methodology**

4.1 Method Overview and Reference Solver

Building on the research gap identified in Section 3, this section defines the hybrid architecture used to test whether a neural force correction can be made dynamically usable in

the unrestricted Three-Body Problem. **SIMON** is a timestep-based integrator: it does not predict the final state directly from the initial state. Instead, it evolves the system step by step using a leapfrog backbone, while using a neural network only as a local scalar correction to the pairwise force magnitude during close encounters.

The reference solution throughout this paper is generated using **REBOUND** [Rein and Liu, 2012], specifically the **ias15** integrator [Rein and Spiegel, 2014]. **ias15** is a high-order adaptive Gauß–Radau method designed for gravitational dynamics and is used here as the benchmark for long-horizon trajectory comparison, boundedness/ejection classification, and runtime comparison. **SIMON** is not intended to replace **ias15** as a universal reference solver. Its purpose is to test a narrower methodological question: which physical constraints must be enforced for a neural-assisted integrator to preserve qualitative long-horizon dynamics?

The method has three core design choices:

1. a second-order leapfrog update provides the integration backbone;
2. Newtonian radial force direction is computed analytically at every step;
3. the neural network learns only a scalar correction to the softened pairwise force magnitude during close encounters.

This separation between learned and analytic components allows the ablation study in Section 5.1 to identify which parts of the method are responsible for stability, accuracy, and computational performance. A compact schematic of the full **SIMON** workflow is provided in Appendix C.

4.2 Scalar Force-Correction Model

For bodies with positions $\mathbf{x}_i$, velocities $\mathbf{v}_i$, and masses m_i , pairwise interactions are computed using the displacement

$$\mathbf{r}_{ij} = \mathbf{x}_j - \mathbf{x}_i, \quad r_{ij} = \|\mathbf{r}_{ij}\|.$$

The exact Newtonian pairwise force is

$$\mathbf{F}_{ij}^{\text{Newt}} = G \frac{m_i m_j}{r_{ij}^3} \mathbf{r}_{ij},$$

where the vector direction is determined entirely by the geometry of the two bodies Murray and Dermott [1999]. This exact radial direction is preserved in **SIMON**; the neural network is never allowed to learn or rotate the force direction.

Close encounters are numerically difficult because the Newtonian force magnitude changes

rapidly as r_{ij} becomes small. To obtain a smooth local correction target, **SIMON** first computes a softened pairwise force using

$$\tilde{r}_{ij} = \sqrt{r_{ij}^2 + \epsilon^2}, \quad \mathbf{F}_{ij}^{\text{soft}} = G \frac{m_i m_j}{\tilde{r}_{ij}^3} \mathbf{r}_{ij}.$$

The network then predicts a scalar correction factor c_{ij} so that

$$\mathbf{F}_{ij}^{\text{SIMON}} = c_{ij} \mathbf{F}_{ij}^{\text{soft}}.$$

Equivalently, the ideal correction satisfies

$$c_{ij} = \left(\frac{\tilde{r}_{ij}}{r_{ij}} \right)^3, \quad \log c_{ij} = 3 \log \left(\frac{\tilde{r}_{ij}}{r_{ij}} \right).$$

The model is therefore trained to predict $\log c_{ij}$ from the feature vector

$$z_{ij} = [\log \tilde{r}_{ij}, \log m_i, \log m_j].$$

This keeps the learned task one-dimensional in the output while preserving the exact Newtonian force direction. The architecture is a small multilayer perceptron,

$$3 \rightarrow 32 \rightarrow 32 \rightarrow 32 \rightarrow 1,$$

with SiLU activations. This compact design is sufficient because the learned function is a smooth scalar correction, not a full vector-valued force law.

4.3 Hybrid Integration and Adaptive Sub-Stepping

At each macro timestep, **SIMON** computes all pairwise forces and updates the state using a velocity-Verlet/leapfrog step. The force calculation is divided into three regimes:

1. **Far field:** if $r_{ij} > 500\epsilon = 0.15 \text{ AU}$, the NN is not invoked and the pairwise force is computed using the exact Newtonian expression.
2. **Close encounter:** if $r_{ij} < 0.15 \text{ AU}$, the NN predicts the scalar correction c_{ij} applied to the softened force.
3. **Safety fallback:** if the encounter is extremely close ($\tilde{r}_{ij} < 5 \times 10^{-4} \text{ AU}$), or if the predicted correction lies outside the gate $0.2 \leq c_{ij} \leq 5.0$, the NN output is rejected and the exact Newtonian force is used instead.

This design follows the principle of hybrid scientific machine learning: learned components are used only where they are intended to help, while physically unsafe outputs are replaced by analytic force evaluations [Ulibarrena et al. \[2024\]](#). In particular, the NN correction

changes only the scalar magnitude of the force; the direction remains the exact radial Newtonian direction throughout.

To resolve close encounters without globally shrinking the timestep, **SIMON** uses adaptive sub-stepping. At each macro step, the minimum pairwise distance $r_{\min}$ is computed. If $r_{\min} < 0.05 \text{ AU}$, the macro step dt is divided into

$$n_{\text{sub}} = \min \left(16, \max \left(2, \left\lceil \frac{0.05}{r_{\min}} \right\rceil \right) \right)$$

sub-steps, each of length dt/n_{sub} . Thus, closer encounters receive finer temporal resolution, while the maximum of 16 sub-steps limits computational overhead. All positions, velocities, accelerations, and forces are stored and updated in float64; the neural-network forward pass uses float32 weights, with the scalar correction converted back to float64 before being applied.

4.4 Training Procedure

Training data is generated from synthetic pairwise configurations sampled across the separation range relevant to the force correction. The target is the analytic correction $\log c_{ij} = 3 \log(\tilde{r}_{ij}/r_{ij})$, so the network learns how to convert the softened force magnitude back toward the Newtonian force magnitude while keeping the force direction analytic. Approximately 60% of samples are drawn from the transition region $r \in [0.5\epsilon, 50\epsilon]$, where softening changes the force appreciably, and 40% from the far field $r \in [50\epsilon, 10]$, where the ideal correction is close to one.

Table 1: Training configuration for the scalar force-correction network.

Component	Setting
Input features	$[\log \tilde{r}_{ij}, \log m_i, \log m_j]$
Target	$\log c_{ij} = 3 \log(\tilde{r}_{ij}/r_{ij})$
Architecture	$3 \rightarrow 32 \rightarrow 32 \rightarrow 32 \rightarrow 1$ with SiLU activations
Training samples	40,000
Validation samples	10,000
Optimizer	AdamW, weight decay 1×10^{-5}
Learning rate schedule	10^{-3} with cosine annealing to 10^{-5} Loshchilov and Hutter [2017]
Loss	Mean squared error on $\log c_{ij}$
Epochs	5,000
Primary training seed	42
Training device	NVIDIA GeForce RTX 5050 CUDA 13.0
Best validation MSE	1.1×10^{-5}

Seed robustness is tested separately in Section 5.3.3 by retraining the same architecture with five independent random initialisations.

4.5 Evaluation Strategy and Scope

Because three-body trajectories are chaotic, long-horizon pointwise agreement alone is not a sufficient evaluation standard. **SIMON** is therefore evaluated using both quantitative and qualitative diagnostics:

Table 2: Evaluation diagnostics used to assess **SIMON**.

Diagnostic	Purpose
Trajectory overlays	Visual comparison of SIMON and ias15 trajectories, including phase drift.
RMS separation	Measures accumulated position deviation from the ias15 reference.
Divergence-rate proxy λ	Summarises early-to-mid horizon separation growth in chaotic rollouts.
Bounded/ejected status	Tests whether SIMON preserves the qualitative dynamical regime.
Energy drift	Checks whether energy error is bounded, timestep-sensitive, and attributable to the integration backbone rather than to the learned correction.
Seed robustness	Tests whether results depend on random NN initialisation.
Speed-accuracy frontier	Compares runtime gain against trajectory error across timesteps.

The fitted value of λ used in this work is a divergence-rate proxy rather than a formal Lyapunov exponent [Bishop, 2025]. It reflects both the intrinsic chaotic sensitivity of the three-body system and the accumulated model/integration mismatch between **SIMON** and **ias15**. A perfect reproduction of the reference trajectory would give $\lambda = 0$ in this comparison, while larger values indicate faster separation from the reference rollout.

The scope of the method is deliberately limited to unrestricted three-body systems and is not intended as a universal replacement for **ias15**. The study pursues three goals: to establish the structural constraints that any neural gravitational integrator must enforce for long-horizon stability; to demonstrate **SIMON** as a reliable characterisation framework for three-body configurations; and to provide a methodology for finding the optimal operating parameters for a given configuration class. The limitations of this scope are discussed further in Section 6.3.

5 Results

In this section we present the experimental results for **SIMON** across four themes. Section 5.1 isolates the individual contribution of each architectural component through a

controlled four-run ablation. Section 5.2 demonstrates **SIMON** as a dynamical characterisation framework across seven initial conditions and 500 randomly generated configurations. Section 5.3 examines physical consistency and reproducibility. Section 5.4 characterises the speed-accuracy trade-off as a separate computational-performance result.

Throughout this section, λ denotes the fitted divergence-rate proxy (slope of $\log \delta(t)$ in the 10–50 yr window), and “bounded” indicates that all three bodies remain within 10 AU of the system barycentre at $T = 100$ yr. The reference integrator is `ias15` [Rein and Spiegel, 2014]; all initial conditions use masses $[1.0, 0.01, 0.005] M_\odot$ unless otherwise stated. At the operational timestep $dt = 0.04$ yr, $\lambda = 0.1655 \text{ yr}^{-1}$ and final RMS = 1.54 AU for the default initial condition (IC1).

5.1 Results I — Stability Decomposition

5.1.1 The Four-Run Ablation Design

To isolate the role of each architectural component, we conduct a controlled four-run ablation in which exactly one component is removed or altered per run while the initial condition IC1, timestep grid, simulation horizon, and evaluation protocol remain fixed. Runs A, C, and D use the same trained **SIMON** weights; Run B uses a separately trained Vector NN baseline because the architecture itself is changed to learn force direction. The four runs are summarised in Table 3.

Table 3: Four-run ablation framework for IC1 over $T = 100$ yr. Each run removes or replaces one component of the full **SIMON** configuration.

Run	Adaptive stepping	Force direction	NN correction	Outcome
A	OFF	Analytic	ON	Ejection (161.7 AU)
B	ON	Learned (Vector NN)	N/A	Ejection (83.0 AU)
C	ON	Analytic	OFF	Bounded (1.67 AU)
D (SIMON)	ON	Analytic	ON	Bounded (1.54 AU)

The logic of the design is as follows. Runs A and B establish stability requirements by showing that removing a component causes ejection. Run C establishes the NN’s role by showing that its removal leaves the system stable. Run D is the full system, providing the accuracy ceiling against which Run C is compared. Because Run B is evaluated on the default initial condition, its trajectory outcome is interpreted together with the gate-sensitivity analysis rather than as a standalone statistical generalisation.

5.1.2 Run A: Adaptive Sub-Stepping Is the Primary Stability Mechanism

Without adaptive sub-stepping, **SIMON** uses a fixed leapfrog step for every integration step regardless of body separation. At $dt = 0.01$ yr, a close encounter occurs whose duration

is shorter than the fixed timestep can resolve. The lightest body receives an incorrectly large gravitational impulse in a single step and is ejected to 161.7 AU at $T = 100$ yr. With adaptive sub-stepping active, the same close encounter is resolved by splitting the step into 2 to 16 sub-steps when $r_{\min} < 0.05$ AU, and ejection is eliminated across all tested timesteps for IC1.

Figure 1 shows the complete frontier sweep across five timestep values: $dt = 0.005, 0.01, 0.02, 0.04$, and 0.08 yr. The sweep is run for both conditions. The worst-case final RMS across the frontier reduces from 161.7 AU before adaptive sub-stepping ($dt = 0.01$ yr) to 1.54 AU after adaptive sub-stepping ($dt = 0.04$ yr), a $105\times$ improvement.

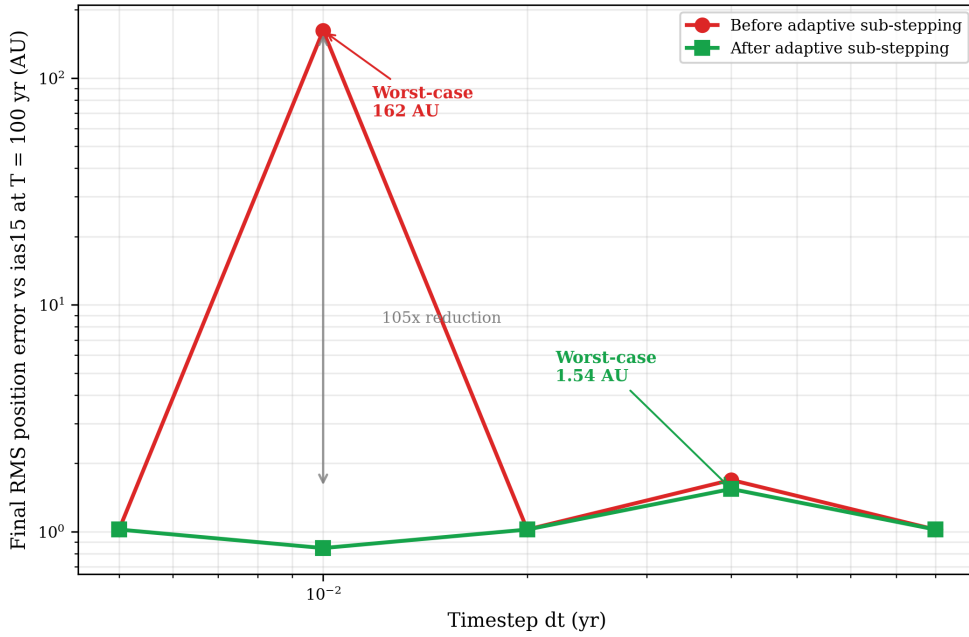


Figure 1: Final RMS position error vs. timestep for SIMON with and without adaptive sub-stepping on IC1.

At the operational timestep $dt = 0.04$ yr—the main timestep used for the operational comparison—adaptive sub-stepping costs +28% per-step overhead ($19.21 \rightarrow 24.57 \mu\text{s}$), reducing speedup from $1.53\times$ to $1.12\times$ relative to *ias15*, while reducing final RMS from 1.691 to 1.543 AU (8.8% improvement) and the divergence-rate proxy from 0.1712 to 0.1655 yr^{-1} (3.3% improvement). Both configurations remain faster than *ias15*. The reported NN invocation rate is less than 1.1% of pair evaluations involving close encounters with $r < 0.15$ AU. Adaptive sub-stepping uses the stricter condition $r_{\min} < 0.05$ AU, so it is triggered only on a subset of those close-encounter cases, leaving the majority of steps at the base leapfrog cost.

5.1.3 Run B: Analytic Force Direction as a Structural Physical Constraint

In Run B, the analytic force direction $\mathbf{u}_{ij} = \mathbf{r}_{ij}/\|\mathbf{r}_{ij}\|$ is replaced by a learned direction from a Vector NN. The Vector NN uses the same hidden-layer structure as **SIMON** but receives strictly more information, including the geometric direction components $[r_x/r, r_y/r, r_z/r]$, and predicts a unit force-direction vector. Post-training verification confirms sub-degree directional accuracy, with maximum directional error approximately 0.04° across the tested distance range. For the tested initial condition, however, the learned-direction architecture produces unbounded trajectories, reaching final RMS error 83.0 AU at $T = 100$ yr, while **SIMON** remains bounded at 1.54 AU. In plain terms, this run tests whether gravity’s direction should be learned approximately or enforced exactly from geometry.

To test whether this unbounded outcome was caused by a single anomalous direction prediction or by accumulated small errors, the Vector NN direction gate was tightened and the simulation was re-run.

Table 4: Gate-sensitivity test for the learned-direction Vector NN baseline. Tightening the direction gate does not prevent the unstable outcome.

Gate setting	Interpretation	Outcome
$\cos(\theta) < 0.866$	Standard 30° gate; triggers only on large direction deviations.	Ejection unchanged; final RMS 83.0 AU.
$\cos(\theta) < 0.9999985$	Much tighter than the standard gate; close to the Vector NN’s measured sub-degree accuracy.	Identical ejection timing; final RMS 83.0 AU.

The gate-sensitivity result shows that the mechanism is not a single catastrophic misprediction, but the accumulation of small non-radial impulses over many steps. When the gate is tightened to $\cos(\theta) < 0.9999985$, corresponding to an angular threshold of approximately 0.1° , the ejection timing and final RMS remain unchanged. This threshold is far tighter than the original 30° gate and close to the Vector NN’s measured sub-degree directional accuracy.

Thus, the failure is not an outlier that a gate can catch. Instead, many individually acceptable direction errors accumulate during close encounters. A simple scale estimate illustrates the issue: at $dt = 0.04$ yr over 100 yr, there are approximately 2,500 integration steps, and 0.04° per close-encounter step can accumulate to roughly 100° of non-radial impulse. Because Newtonian gravity is a central force directed exactly along the line connecting two bodies, any learned perpendicular component has no physical basis and can compound into non-physical energy injection in a chaotic system.

Figures 2 and 3 illustrate the physical consequence of the gate-sensitivity finding. During

the bounded phase, *SIMON* and the Vector NN have nearly identical fitted divergence rates (0.1655 vs. 0.1656 yr^{-1}), but the learned-direction model later becomes unbounded for this tested initial condition, reaching final RMS error 83.0 AU . In Figure 3, the long red dotted segments are escape paths of the Vector NN trajectory, not stable orbital loops.

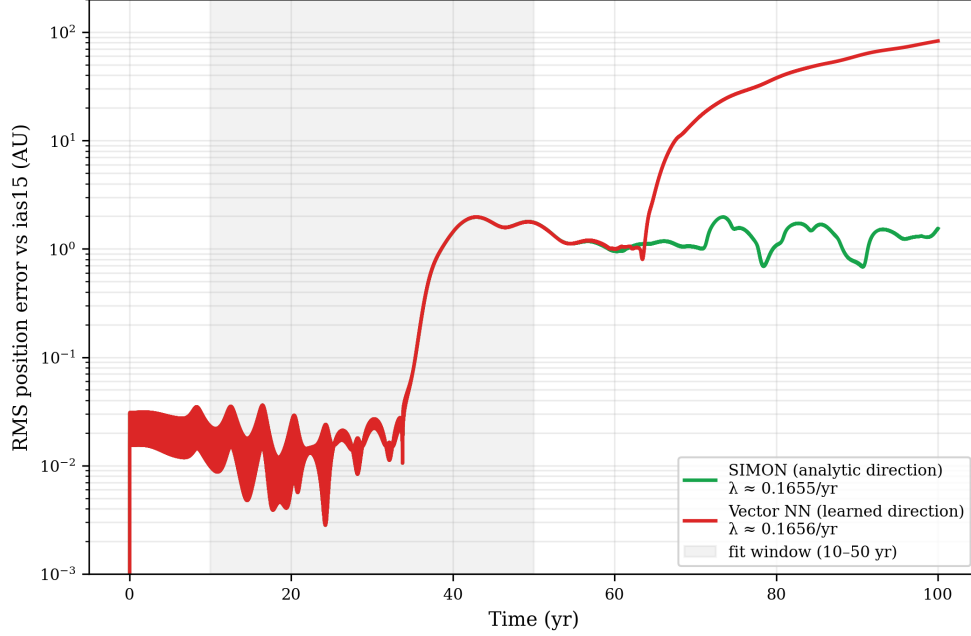


Figure 2: RMS position error vs. time for *SIMON* and the Vector NN baseline. The shaded region marks the 10–50 yr divergence-rate fit window.

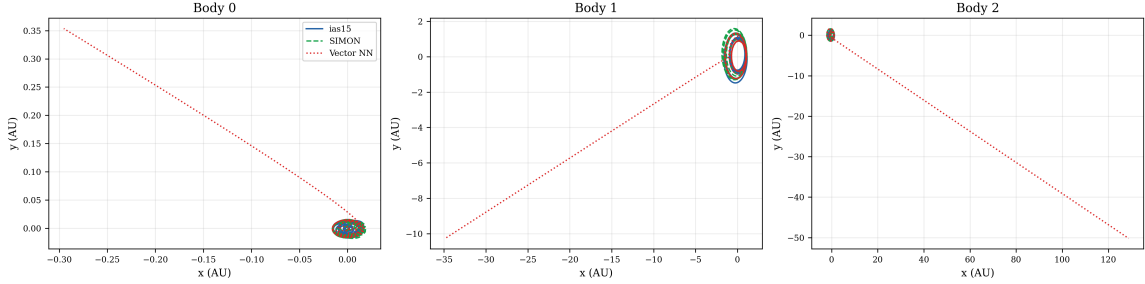


Figure 3: XY trajectory overlays for *ias15*, *SIMON*, and the Vector NN over 100 years. Each body is shown separately with independent axis limits.

This result supports the conclusion that analytic force direction is not merely a convenience but a structural stability constraint. For the tested initial condition, the learned-direction baseline becomes unbounded despite sub-degree angular accuracy, and the gate-sensitivity analysis shows that the outcome is insensitive to the specific gate threshold chosen. Analytic direction therefore removes, by construction, a non-physical perpendicular-force error mode that a learned direction model can still introduce during close encounters.

5.1.4 Run C: The Neural Network Correction Is an Accuracy Mechanism

In Run C, the NN correction is disabled entirely: the correction factor c is fixed at 1.0 for all pair evaluations, reducing the force to pure softened Newtonian gravity. The analytic force direction and adaptive sub-stepping remain active. The result is a bounded simulation at 1.67 AU—not an ejection.

Figure 4 shows that Run C (No-NN baseline, red dashed) and Run D (full SIMON, green solid) are visually indistinguishable throughout the 100-year horizon. Both achieve $\lambda = 0.1655 \text{ yr}^{-1}$ to four decimal places. The NN correction contributes 8.5% improvement in final RMS accuracy ($1.67 \rightarrow 1.54 \text{ AU}$) at the 0.45% of pair evaluations involving close encounters below 0.15 AU, where it is invoked.

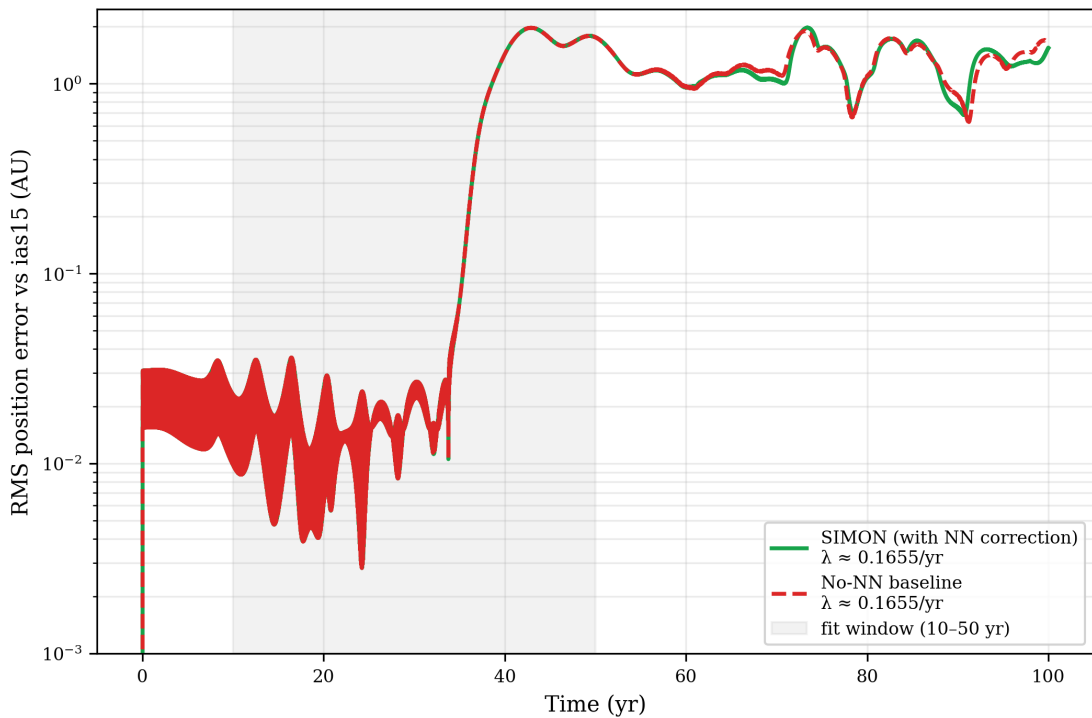


Figure 4: RMS position error vs. time for full SIMON and the No-NN baseline. Both runs retain analytic direction and adaptive sub-stepping.

This result clarifies the role of the NN correction. Run A shows that close-encounter stability is provided by adaptive sub-stepping, since removing that mechanism causes ejection even when the NN correction remains active. Run C shows that removing the NN correction while retaining adaptive sub-stepping leaves the system bounded, but increases final RMS error from 1.54 AU to 1.67 AU. Therefore, the NN correction is best interpreted as a targeted accuracy refinement at close-encounter steps, rather than the primary stability mechanism.

5.1.5 Synthesis: The Two-Constraint Principle

The four-run ablation establishes a clear decomposition of **SIMON**'s performance:

- **Adaptive sub-stepping** eliminates the leapfrog ejection failure at $dt = 0.01$ yr for IC1 (Run A: 161.7 AU without sub-stepping versus 0.847 AU with sub-stepping).
- **Analytic force direction** is a structural physical constraint (Run B: for the tested initial condition, replacing analytic direction with a learned direction produces unbounded trajectories with final RMS 83.0 AU).
- **NN correction** is an accuracy mechanism (Run C: removing it leaves stability intact but costs 8.5% accuracy).
- **Full SIMON** (Run D) achieves the best accuracy by combining all three components.

Figures 5 and 6 show the XY trajectory overlay and coordinate time series for the full **SIMON** system (Run D), confirming bounded dynamics for all three bodies over 100 years.

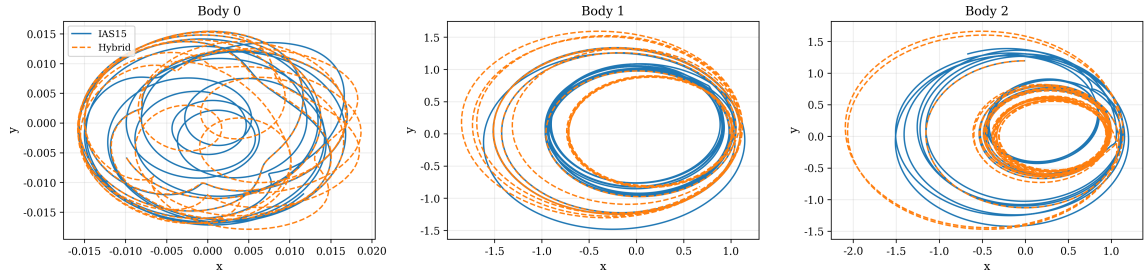


Figure 5: XY trajectory overlays for all three bodies comparing **SIMON** with **ias15** over 100 years.

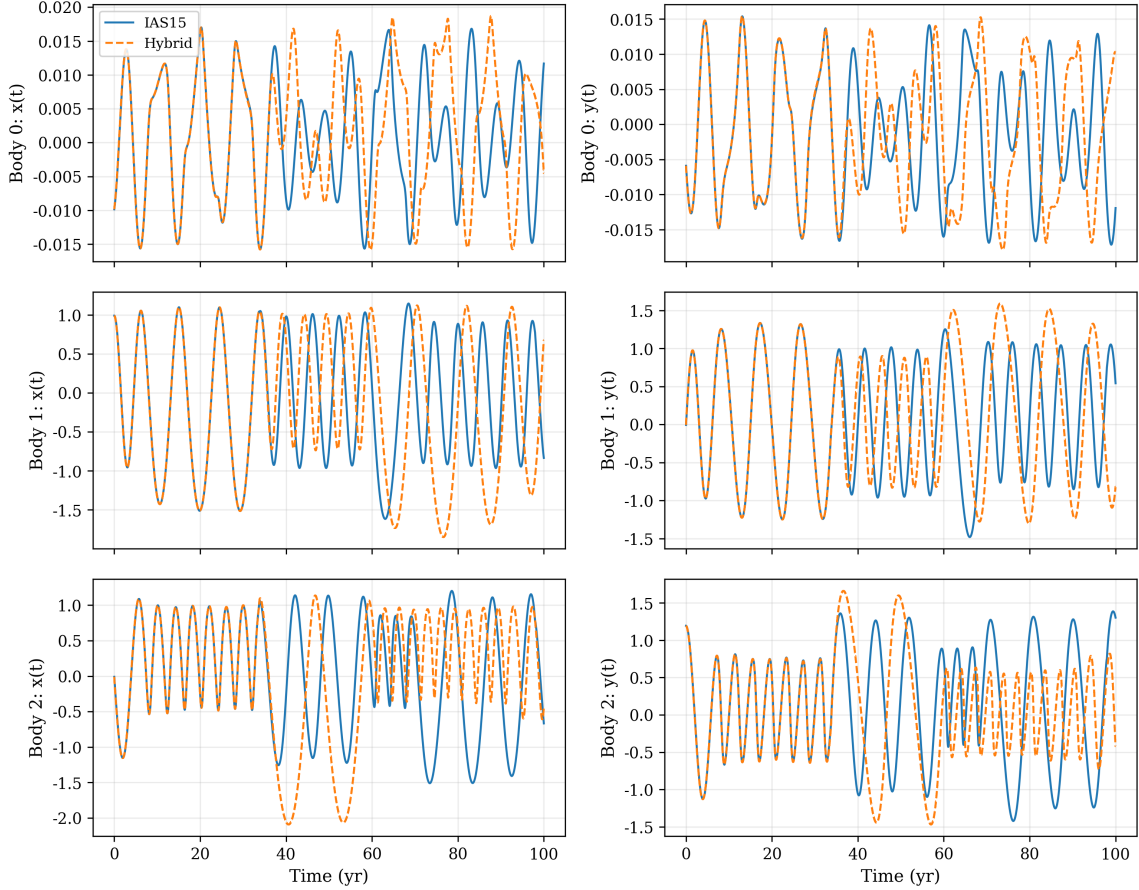


Figure 6: Coordinate time series $x(t)$ and $y(t)$ for all three bodies over 100 years. Phase drift accumulates while the trajectories remain bounded.

The primary error mode is phase drift—a gradual timing mismatch that accumulates while preserving the qualitative orbital geometry and boundedness of all three bodies. This is consistent with the chaotic sensitivity of the Three-Body Problem [Ott, 2002] and does not indicate structural failure of the integrator.

5.2 Results II — Generalisation Across Initial Conditions

5.2.1 Seven Initial Conditions Spanning the Dynamical Spectrum

Having established the two-constraint principle, we demonstrate **SIMON** as a characterisation framework across seven distinct dynamical regimes. A researcher brings their three-body configurations to **SIMON** and receives bounded/ejected classification, divergence rates, and trajectory behaviour — the outputs needed to characterise the dynamical properties of the configuration. Table 5 summarises outcomes for all seven ICs. All runs use masses $[1.0, 0.01, 0.005] M_{\odot}$ except IC2 (near-equal mass), $T = 100\text{yr}$, and $dt = 0.04\text{yr}$.

Table 5: Multi-IC evaluation results at $T = 100$ yr and $dt = 0.04$ yr. Summary statistics are computed over the four bounded configurations only.

IC	Regime	λ (yr $^{-1}$)	RMS (AU)	Bounded	Speedup
IC1: Default	Moderate scattering	0.1655	1.54	Yes	1.12×
IC2: Near-equal	Strongly interacting	0.0430	116.9	No	1.77×
IC3: Tight pair	High encounter rate	0.1207	3.45	Yes	1.24×
IC4: Hierarchical	Weakly coupled outer	0.0990	1.60	Yes	1.14×
IC5: High ecc. (v1)	High eccentricity	0.0648	20.0	No	1.22×
IC6: Near-circular	Low chaos	−0.019	0.008	Yes	1.08×
IC7: High ecc. (v2)	High eccentricity	0.0341	11.4	No	1.11×
Mean (bounded ICs only)		0.092	1.65	—	—
Std (bounded ICs only)		0.068	1.22	—	—
Bounded count		—	—	4/7	—

5.2.2 Bounded Configurations: Four ICs

All four stable configurations maintain bounded dynamics for the full 100-year horizon. The configurations span the complete stable dynamical spectrum of the three-body problem:

- **IC6 (near-circular):** $\lambda \approx 0$, final RMS = 0.008 AU. The NN is never invoked (NN_frac = 0.000); **SIMON** computes exact Newtonian forces for every step. The residual 0.008 AU is the pure leapfrog–Gauss–Radau integrator difference.
- **IC4 (hierarchical):** $\lambda = 0.099 \text{ yr}^{-1}$, final RMS = 1.60 AU. The outer body at 5 AU is weakly coupled; the NN is also never invoked.
- **IC1 (default):** $\lambda = 0.1655 \text{ yr}^{-1}$, final RMS = 1.54 AU. The default IC1 baseline.
- **IC3 (tight binary):** $\lambda = 0.121 \text{ yr}^{-1}$, final RMS = 3.45 AU. The inner body at 0.5 AU generates continuous close encounters; despite this, **SIMON** remains bounded throughout.

Figure 7 summarises the divergence rates, final RMS values, and speedups across all seven ICs.

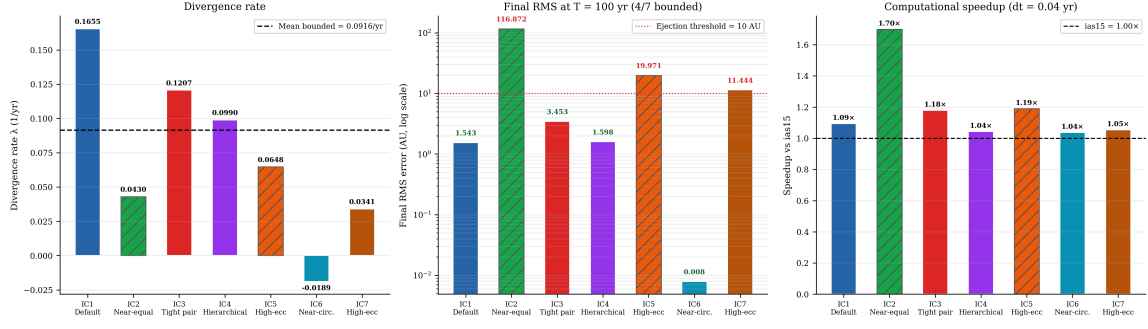


Figure 7: Summary across seven initial conditions: divergence rate, final RMS error, and speedup relative to *ias15*. Hatched bars denote physically unstable configurations.

Figure 8 shows the divergence curves for all seven ICs simultaneously.

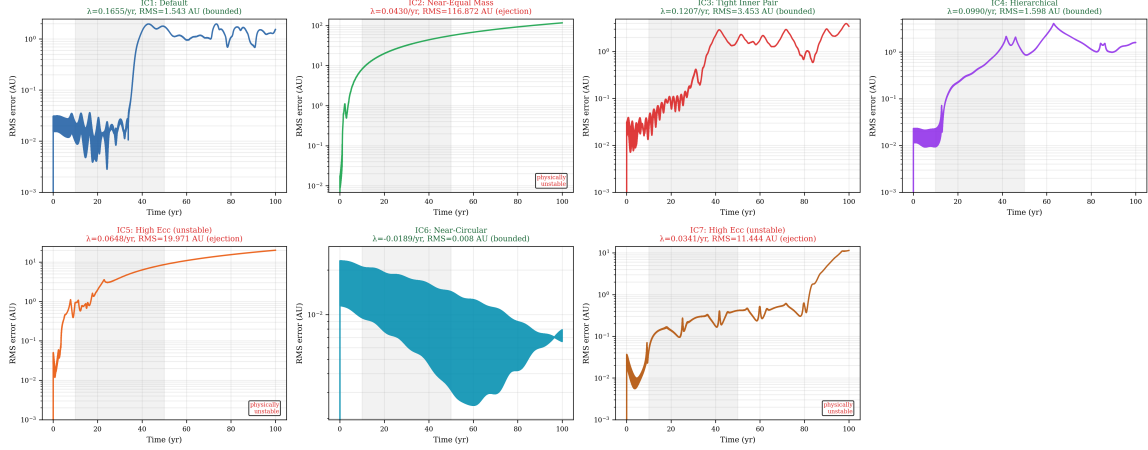


Figure 8: RMS position error vs. time for all seven initial conditions. The shaded region marks the 10–50 yr divergence-rate fit window.

5.2.3 Physically Unstable Configurations

Three configurations produce ejections (IC2, IC5, IC7). In each case, the reference integrator *ias15* independently also ejects a body, confirming these are physical instabilities rather than SIMON failures:

- **IC2 (near-equal mass):** SIMON ejects to 116.9 AU; *ias15* ejects to 58.9 AU. The $2\times$ post-ejection discrepancy is consistent with chaotic amplification of integrator differences after the system becomes dynamically unbound.
- **IC5 (high eccentricity, $v = 2.2$, $r = 0.3$ AU):** Both integrators eject; *ias15* ejects Body 2 at $t = 34.8$ yr.
- **IC7 (high eccentricity redesign, $v = 1.6$, $r = 0.6$ AU):** Both integrators eject; *ias15* ejects Body 2 at $t = 88.4$ yr.

SIMON does not suppress these physical instabilities. This is expected and physically consistent: the model agrees with **ias15** on the occurrence of dynamical instability rather than artificially stabilising configurations that are physically unbound. The post-ejection distances are not expected to match exactly because chaotic divergence amplifies integrator differences after unbinding. High-eccentricity configurations with these mass ratios proved intrinsically unstable in both cases tested: the eccentric inner body transfers orbital energy to the outer body through repeated periapsis passages, eventually unbinding it regardless of integrator.

5.2.4 Population-Scale Generalisation Validation

To confirm that **SIMON**'s qualitative generalisation is not an artefact of the seven hand-selected configurations, we additionally evaluated both **SIMON** and **ias15** on 500 randomly generated three-body configurations. The population was restricted to filtered hierarchical systems: masses sampled log-uniformly from the scalar network's training range, inner periapsis above 0.20 AU, and the outer body constrained to remain non-crossing ($q_{\text{out}} > 1.5 r_1$), so that all configurations lie within **SIMON**'s validated operating regime. At $dt = 0.08 \text{ yr}$, **SIMON** and **ias15** agreed on the bounded/ejected classification for 96.0% of configurations (480/500). **ias15** classified 112/500 systems as ejected; **SIMON** classified 124/500 as ejected, with 16 false ejections (3.2%) and 4 false bounded classifications (0.8%). For the 372 configurations bounded under both integrators, the mean divergence rate was $\lambda = 0.023 \pm 0.021 \text{ yr}^{-1}$, below the seven-IC mean of 0.092 yr^{-1} as expected from the hierarchical filtering, but within the range observed across all seven dynamical regimes (-0.019 to 0.166 yr^{-1}). This confirms that **SIMON**'s bounded/ejected agreement is not limited to the hand-selected configurations.

5.3 Results III — Physical Consistency and Reproducibility

5.3.1 Energy Conservation

We track fractional energy drift

$$\frac{\Delta E(t)}{E_0} = \frac{E(t) - E_0}{|E_0|},$$

where $E(t) = \text{KE}(t) + \text{PE}(t)$ is the total mechanical energy. This test has three purposes: to verify that energy drift is not caused by the learned correction; to interpret IC4's larger drift value; and to characterise how energy conservation depends on timestep and initial condition. Table 6 reports the maximum absolute fractional drift over $T = 100 \text{ yr}$ for three representative bounded initial conditions across seven timestep values.

Table 6: Maximum fractional energy drift $|\Delta E/E_0|_{\max}$ for **SIMON** over $T = 100$ yr. The main experiments use $dt = 0.04$ yr.

Initial condition	$ E_0 $	NN frac.	$dt = 0.01$	$dt = 0.02$	$dt = 0.04$	$dt = 0.05$	$dt = 0.06$	$dt = 0.08$	$dt = 0.10$
IC1 Default	0.0072	0.0045	0.22%	1.30%	1.39%	0.34%	0.38%	1.86%	1.03%
IC3 Tight Pair	0.0133	0.0000	0.02%	0.27%	0.67%	0.65%	0.82%	1.80%	2.52%
IC4 Hierarchical	0.0060	0.0000	1.05%	3.89%	13.77%	22.81%	24.11%	25.18%	32.28%

Attribution. The energy drift is not caused by the neural correction. IC3 and IC4 both have NN fraction = 0.0000, so the learned force correction is never invoked during those rollouts. Their nonzero energy drift must therefore arise from the fixed-timestep leapfrog backbone rather than from NN-driven force errors.

IC4 interpretation. IC4’s higher fractional drift at the operational timestep (13.77% at $dt = 0.04$ yr) reflects its small binding energy $|E_0| = 0.0060$ — the smallest of the three configurations tested. Because fractional drift is normalised by $|E_0|$, weakly bound hierarchical configurations show larger relative energy drift at the same timestep. This is not NN-driven energy injection: the NN is never invoked for IC4. The rollout remains fully bounded over $T = 100$ yr, consistent with the multi-IC result (final RMS = 1.60 AU).

Timeseries character. Figure 9 shows the IC4 energy-drift time series across all seven tested timestep values. At $dt = 0.04$ yr (orange), the drift is oscillatory rather than monotonically increasing, consistent with bounded finite-timestep energy error from a leapfrog-type backbone. At $dt = 0.05$ yr and coarser, the time series shows a sharp energy-baseline elevation near $t \approx 10$ yr: the orbit shifts to a higher-energy configuration from that point and the integration continues on the perturbed trajectory. The three values at $dt = 0.05$, 0.06 , and 0.08 yr — 22.81%, 24.11%, and 25.18% — cluster tightly together, suggesting that the dominant error is associated with the same under-resolved encounter rather than gradual accumulated drift. The threshold for IC4 lies between $dt = 0.04$ and $dt = 0.05$ yr. The choice of $dt = 0.04$ yr as the operational timestep is therefore a conservative design decision for this particular hierarchical stress case, not a universal requirement.

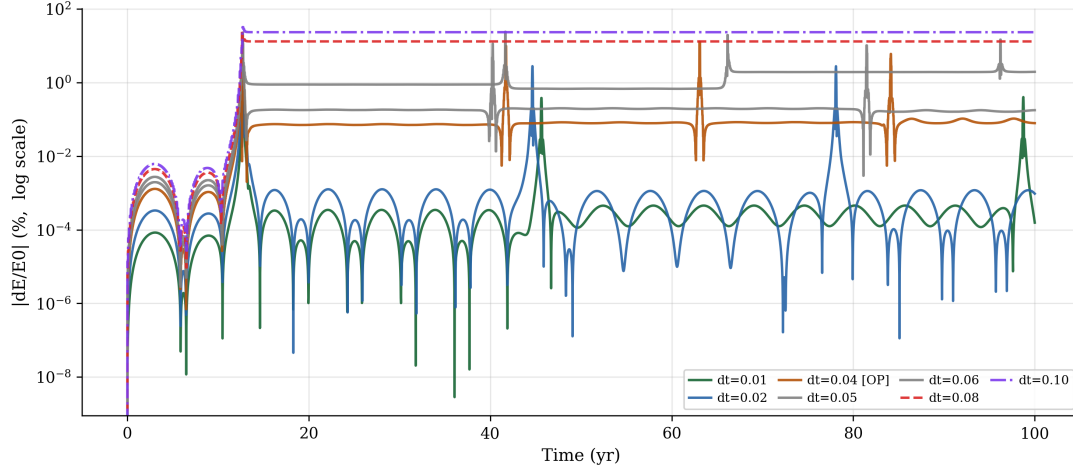


Figure 9: IC4 energy drift across timestep values over $T = 100$ yr. Drift rises sharply above $dt = 0.04$ yr.

5.3.2 Timestep Sensitivity of Energy Drift

Table 6 shows that energy drift depends on both timestep and initial condition, and the two interact differently across the dynamical spectrum.

IC3: smooth scaling. IC3 gives the clearest monotonic trend, rising from 0.02% at $dt = 0.01$ yr to 2.52% at $dt = 0.10$ yr with no exceptions. This is consistent with second-order leapfrog accumulation: a tight-pair configuration with frequent but well-resolved encounters produces energy error that scales predictably with timestep.

IC4: threshold behaviour. IC4 is the most timestep-sensitive case. Below the threshold, the drift shows second-order-like timestep dependence: reducing the timestep from $dt = 0.04$ to $dt = 0.01$ yr lowers the maximum drift from 13.77% to 1.05%, a $13.2\times$ reduction for a $4\times$ finer timestep. Above the threshold, the pattern changes entirely: the three values at $dt = 0.05, 0.06$, and 0.08 yr differ by less than 3% absolute despite a 60% increase in timestep, reflecting a single encounter failure rather than gradual accumulation.

IC1: configuration-dependent behaviour. IC1 behaves differently from both IC3 and IC4. Its drift remains below 2% across all seven tested timesteps and is non-monotonic: the values do not consistently increase with timestep. This reflects the fact that IC1’s energy error at these small magnitudes is dominated by chaotic trajectory differences between dt values rather than by systematic integration error. For this moderate-scattering configuration, operating at coarser timesteps is a credible design choice with no meaningful energy penalty.

SIMON as a configurable framework. These results show that **SIMON**’s energy conservation properties are determined jointly by the timestep and the dynamical regime of the initial condition. The appropriate operating point is not a single universal value but a design choice that should be determined against representative initial conditions for the target problem. The two-step methodology in Section 5.4.1 provides a principled basis for making that selection.

5.3.3 Training Seed Robustness

To verify that results from the seed-42 model used throughout this paper are not specific to a particular random initialisation, training was repeated five times with seeds {42, 123, 456, 789, 1000}. Table 7 summarises the training and simulation outcomes.

Table 7: Training seed robustness for the full **SIMON** run on IC1 at $T = 100$ yr and $dt = 0.04$ yr.

Seed	Val. MSE	λ (yr ⁻¹)	Bounded
42 (primary model)	1.11×10^{-5}	0.1655	Yes
123	1.72×10^{-5}	0.1655	Yes
456 (worst)	1.75×10^{-5}	0.1655	Yes
789	1.68×10^{-5}	0.1655	Yes
1000	1.43×10^{-5}	0.1655	Yes
Mean	1.54×10^{-5}	0.1655	5/5
CV	15.8%	0%	—

The divergence rate $\lambda = 0.1655 \text{ yr}^{-1}$ is identical to four decimal places across all five seeds. All five independently trained models produce bounded dynamics with identical fitted divergence rate, confirming that downstream stability is insensitive to random initialisation.

The 15.8% coefficient of variation in validation MSE does not imply dynamical sensitivity to seed choice: the variation is concentrated at sub-softening distances ($r < 5 \times 10^{-4}$ AU) where the safety gate bypasses NN predictions entirely. In the operationally relevant range ($r \geq 10^{-3}$ AU), all seeds produce correction factor errors below 0.3%—functionally identical—and the simulation metrics are invariant.

Figure 10 shows the training convergence and per-seed validation MSE.

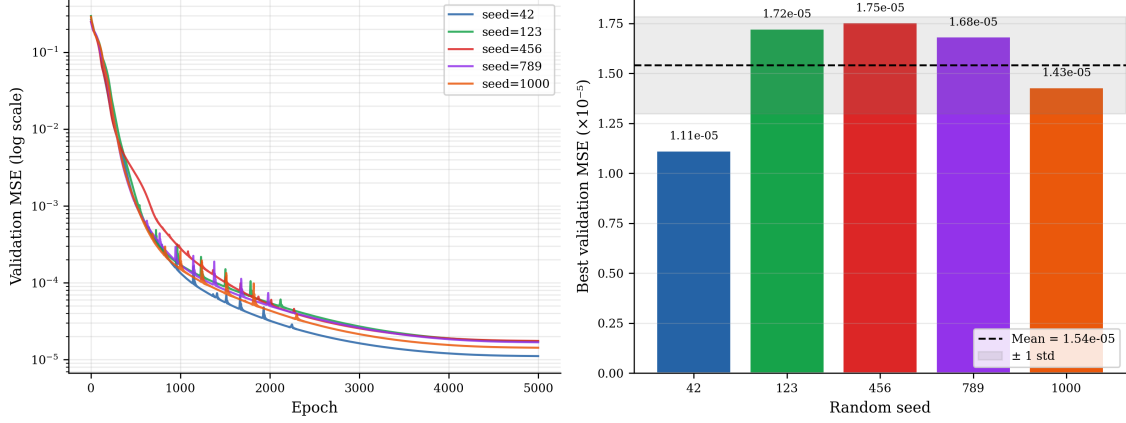


Figure 10: Training seed robustness across five random initialisations. Left: validation MSE curves; right: best validation MSE per seed.

5.4 Results IV — Computational Performance

5.4.1 Speed–Accuracy Frontier on IC1

The speed–accuracy frontier measures the trade-off between simulation throughput and trajectory agreement with `ias15` across five tested timestep values, using IC1 (the default initial condition) as the benchmark configuration.

Accuracy metric. Previous presentations of this frontier used final RMS position error at $T = 100$ yr as the accuracy axis. In a chaotic three-body system, however, any two integrations diverge exponentially past the Lyapunov time (~ 6 yr for IC1), so a single endpoint snapshot at $T = 100$ yr is effectively a random sample from the chaotic attractor rather than a reliable measure of integration quality. A coarser timestep may appear to give lower final RMS simply because its chaotic trajectory happened to lie closer to the `ias15` trajectory at that specific instant.

To reduce this single-snapshot sensitivity, we replace the endpoint metric with the *time-averaged RMS deviation* over the full 100-year rollout:

$$\overline{\text{RMS}}_{0:T} = \sqrt{\frac{1}{N_t} \sum_{k=1}^{N_t} \frac{1}{3} \sum_{i=1}^3 \|\mathbf{x}_{i,\text{SIMON}}(t_k) - \mathbf{x}_{i,\text{ias15}}(t_k)\|^2},$$

where the sum is over all $N_t = 5000$ equally spaced sample points from $t = 0$ to $T = 100$ yr. This metric uses the full trajectory, includes the pre-chaotic window where the comparison is most physically meaningful, and is harder to distort by a single lucky or unlucky endpoint.

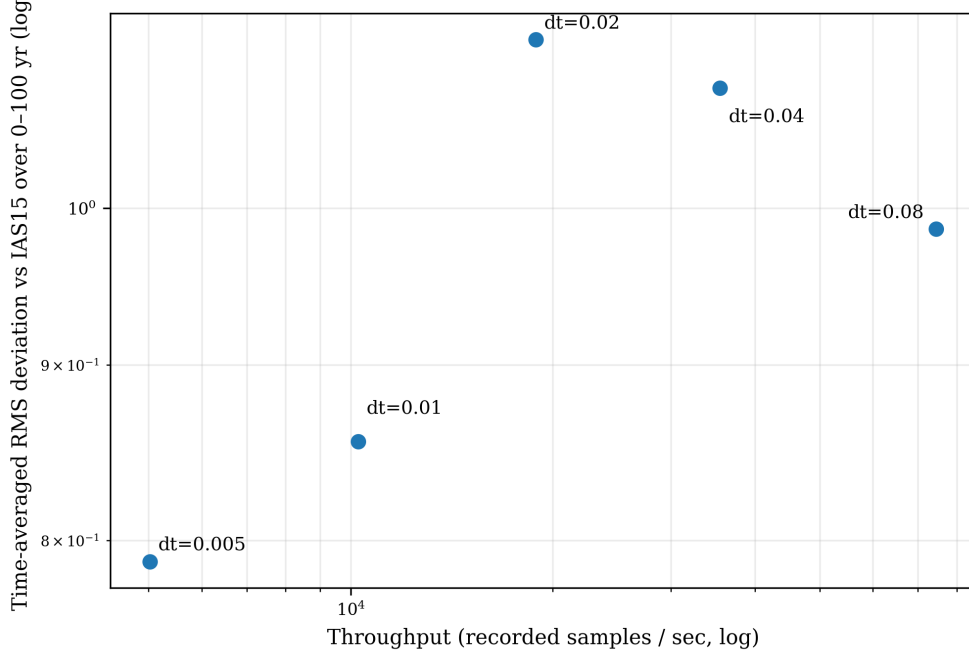


Figure 11: Speed-accuracy frontier for **SIMON** on IC1. x : throughput (samp/s); y : time-averaged RMS vs **ias15**. Lower-right is better.

Frontier results. Table 8 gives the full quantitative results. The **ias15** baseline completes in 0.080 s.

Table 8: Speed-accuracy frontier results for **SIMON** on IC1 over $T = 100$ yr. Accuracy is time-averaged RMS deviation from **ias15**.

dt (yr)	Time-avg RMS (AU)	Throughput (samp/s)	Time/step (μ s)	Total time (s)	Steps	Speedup
0.005	0.789	9,285	26.9	0.538	20,000	$0.15\times$
0.010	0.855	18,307	27.3	0.273	10,000	$0.29\times$
0.020	1.120	35,672	28.0	0.140	5,000	$0.57\times$
0.040	1.084	69,880	28.6	0.072	2,500	$1.12\times$
0.080	0.986	135,629	29.5	0.037	1,250	$2.18\times$

Speed observations. The per-step time ranges from 26.9μ s at $dt = 0.005$ yr to 29.5μ s at $dt = 0.08$ yr: a range of only 2.6μ s across a $16\times$ change in timestep. This nearly flat profile indicates that the cost of an individual **SIMON** step is approximately constant and is dominated by force evaluation, pairwise distance checks, and occasional adaptive sub-stepping overhead. The observed runtime improvement at larger dt therefore comes primarily from taking fewer macro steps, not from making individual steps cheaper. For a fixed $T = 100$ yr horizon, the number of macro steps scales as T/dt , so total runtime scales roughly as (T/dt) times the near-constant per-step cost.

Accuracy observations. Under the time-averaged metric, the frontier reveals a clearer picture than the endpoint metric alone. The $dt = 0.005$ and $dt = 0.01$ yr settings achieve

the lowest time-averaged deviation (0.79 and 0.86 AU respectively) but run $3\text{--}7\times$ slower than `ias15`. The $dt = 0.02$ yr setting produces the worst overall combination: slower than `ias15` and with the highest time-averaged deviation (1.12 AU), reflecting earlier trajectory divergence that accumulates across the full rollout. At $dt = 0.04$ yr, `SIMON` achieves $1.12\times$ speedup with a time-averaged deviation of 1.08 AU. At $dt = 0.08$ yr, it achieves $2.18\times$ speedup with a time-averaged deviation of 0.99 AU — lower than $dt = 0.04$ yr, confirming that this result is not a lucky endpoint artefact but reflects consistently closer trajectory agreement over the full rollout.

Two-step parameter optimisation methodology. The energy-drift sweep in Section 5.3.2 and the speed–accuracy frontier above together constitute a methodology for finding the optimal operating timestep for any target three-body configuration. In the first step, the energy-drift sweep defines the feasible dt range: values for which energy conservation shows physically valid, oscillatory shadow-Hamiltonian behaviour for the target configuration. In the second step, the frontier reveals the trade-off between computational speedup and time-averaged trajectory accuracy within that feasible range, allowing the user to select their optimal operating point. For IC1, the energy sweep confirms $dt = 0.08$ yr as feasible (1.86% drift, oscillatory behaviour); the frontier shows this yields approximately $2.2\times$ speedup over `ias15`. For IC4, the energy sweep identifies $dt = 0.04$ yr as the threshold for valid energy behaviour; the frontier places this at approximately $1.1\times$ speedup. `SIMON`’s operational speedup is therefore a configuration-dependent range rather than a single number, spanning approximately $1.1\times$ to $2.2\times$ across the tested dynamical spectrum.

6 Discussion and Conclusion

6.1 The Two-Constraint Principle

The central finding of this work is that long-horizon orbital stability in a hybrid neural three-body integrator is not produced by the neural network alone. It requires two complementary physical constraints: adaptive sub-stepping and analytic force direction. Adaptive sub-stepping operates at the integration level by resolving close encounters that a fixed macro-timestep can under-resolve. Analytic force direction operates at the force-computation level by enforcing the exact Newtonian radial symmetry at every step.

The ablation study shows that these constraints have distinct roles. Removing adaptive sub-stepping causes an under-resolved close encounter to produce ejection in the timestep frontier. Replacing analytic force direction with a learned direction also leads to unbounded motion for the tested initial condition. By contrast, removing the neural correction while retaining both physical constraints preserves bounded dynamics, with

only a modest increase in final RMS error. Thus, the neural network correction is best understood as an accuracy refinement, while adaptive sub-stepping and analytic direction are the main stability mechanisms.

The broader implication is that neural models for chaotic physical systems should not freely learn quantities whose symmetries are already known exactly. In **SIMON**, Newtonian radial direction is enforced analytically, and the network is restricted to a low-dimensional scalar correction during close encounters. This constitutes a general design principle for neural gravitational integration: preserve exact physical structure where available, and use learning only for targeted corrections where analytic or numerical approximations are most fragile. The principle applies to any neural force-correction integrator in a chaotic gravitational system, not only to the specific architecture studied here.

6.2 Positioning **SIMON** Relative to **ias15**

SIMON should not be interpreted as a replacement for **ias15**. The reference integrator remains substantially more accurate in energy conservation because it uses high-order adaptive Gauß–Radau integration. **SIMON**, by contrast, uses a second-order leapfrog backbone with targeted sub-stepping, so its energy drift is larger and follows the expected timestep-dependent behaviour of the underlying integrator.

The relevant comparison is therefore not whether **SIMON** reproduces every **ias15** coordinate exactly over 100 years. In a chaotic three-body system, small model or numerical differences naturally accumulate as phase drift. The more meaningful question is whether the hybrid method preserves the qualitative dynamical regime: bounded configurations should remain bounded, physically unstable configurations should be identified as unstable, and energy drift should remain physically interpretable.

Under this interpretation, **SIMON** serves two further roles beyond proving the two-constraint principle. As a characterisation framework (Section 5.2), it reliably classifies bounded and ejected configurations, delivering bounded/ejected status, divergence rates, and trajectory behaviour across the dynamical spectrum — confirmed at 96% agreement with **ias15** across 500 randomly generated configurations. As an optimisation tool (Section 5.4), the two-step parameter methodology finds the maximum achievable speedup for any given configuration: the energy-drift sweep identifies the feasible dt range, and the speed–accuracy frontier reveals the optimal operating point within that range. For IC1 this yields approximately $2.2\times$ speedup at $dt = 0.08$ yr; for IC4, $dt = 0.04$ yr is the energy threshold, giving approximately $1.1\times$ speedup.

The same interpretation is supported beyond the synthetic initial conditions used in the main ablations. [Appendix A](#) tests **SIMON** on the real Sun–Earth–Moon system using NASA/JPL Horizons state vectors, while [Appendix B](#) verifies locally that the learned

scalar correction reduces close-range force- magnitude error in its intended operating regime. These checks do not change the main conclusion; they strengthen its physical interpretation by connecting the controlled ablations to a real astronomical configuration and to the force-level role of the NN.

To answer the research question:

What structural constraints must a neural force-correction integrator enforce to achieve long-horizon orbital stability in a chaotic three-body system, and how can such an integrator be used to find the maximum achievable speedup over traditional solvers for a given dynamical configuration?

In the tested three-body regimes, two structurally enforced physical constraints are necessary and sufficient for long-horizon stability; the neural correction contributes accuracy but not stability. Built on this principle, **SIMON** reliably characterises three-body configurations across the dynamical spectrum (96% agreement across 500 randomly generated configurations), and the two-step methodology finds $1.1\times$ to $2.2\times$ speedup across the tested configurations. The main conclusion is therefore not that **SIMON** replaces **ias15**, but that a neural force-correction integrator satisfying the two-constraint principle can reliably characterise three-body dynamics and systematically find the optimal operating parameters for any given configuration class.

6.3 Limitations and Future Work

This work should be interpreted as a controlled methodological study rather than as a universal neural replacement for high-order reference integrators. Several limitations define the scope of the present results and suggest natural directions for future work.

First, the ablation study in Section 5.1 is centred on IC1. Although the multi-IC evaluation in Section 5.2 provides evidence that the full **SIMON** configuration generalises across several regimes, the causal decomposition of adaptive sub-stepping, analytic direction, and neural correction should be repeated over a broader set of mass ratios and initial geometries.

Second, the fitted divergence-rate λ is a proxy rather than a formal Lyapunov exponent. It compares **SIMON** against **ias15**, so it reflects both intrinsic chaotic divergence and accumulated model mismatch. Future work could compute formal Lyapunov exponents by evolving nearby initial conditions under the same dynamics and comparing them with the model-to-reference divergence used here [Bishop, 2025].

Third, the operating timestep and close-encounter thresholds are tuned for the tested three-body systems. Larger systems, different mass ratios, or more extreme eccentricities may require different thresholds, broader training distributions, or additional safeguards.

The pairwise structure of the correction model should extend naturally to $N > 3$ systems, but this has not yet been demonstrated.

Fourth, the present implementation is not optimised for maximum computational speed. The two-step methodology identifies an achievable speedup range of $1.1\times$ to $2.2\times$ across the tested configurations, but this range is constrained by runtime dominated by force evaluation, gating, and sub-stepping overhead rather than integration arithmetic. Future implementations could extend the achievable range through vectorised pair evaluation, batching, GPU inference, or lower-level compiled kernels.

SIMON demonstrates that learned force correction can be useful in chaotic gravitational simulation when the physically dangerous parts of the problem—force direction and close-encounter resolution—are constrained analytically. Built on this foundation, it further demonstrates that a trustworthy hybrid neural integrator can reliably characterise three-body configurations across the dynamical spectrum and systematically find the maximum achievable speedup for any given configuration class.

Appendix A: Physical Validation: Sun–Earth–Moon

To address the limitation that the main experiments use synthetic three-body initial conditions, we tested SIMON on the real Sun–Earth–Moon system using NASA/JPL Horizons state vectors. We performed two validation checks. First, we initialized the system from a present-day Horizons state and ran a forward 100-year timestep-frontier test. Second, we initialized the system from a historical Horizons state in 1926 and compared the century-scale rollout against both Horizons ephemeris samples and a simplified three-body `ias15` reference.

These experiments are not intended to reproduce the full Solar System ephemeris, since other planets are omitted. Instead, they test whether SIMON preserves the physically important Sun–Earth and Earth–Moon distance structure when initialized from real astronomical data.

A.1 Forward 100-Year Timestep-Frontier Test

For the forward validation, initial positions and velocities were obtained from NASA/JPL Horizons at epoch 2026-01-01, converted to AU, years, and solar masses, and shifted into the centre-of-mass frame. In these units, $G = 4\pi^2$. The reference solution was generated using `ias15`. SIMON was run for $T = 100$ yr at $dt = 0.008, 0.012, 0.016$ yr, both with and without adaptive sub-stepping.

Because the Moon can become unbound from Earth while the full system remains within 10 AU of the centre of mass, we report both the paper’s global boundedness criterion and a lunar-bound-state outcome based on the Earth–Moon distance range.

Table 9: Forward Sun–Earth–Moon validation over $T = 100$ yr.

dt	Mode	$\text{RMS}_{0:T}$ (AU)	$E_{\max}$ (%)	Earth–Moon range (AU)	Outcome
0.008	ON	2.31×10^{-2}	1.71×10^{-5}	0.002376–0.002644	Valid
0.008	OFF	2.41×10^{-1}	4.42×10^{-3}	0.002360–0.003357	Degraded
0.012	ON	3.40×10^{-2}	3.78×10^{-5}	0.002376–0.002647	Valid
0.012	OFF	9.74×10^{-1}	1.11×10^{-2}	0.002190–1.986214	Lunar orbit lost
0.016	ON	4.45×10^{-2}	6.64×10^{-5}	0.002378–0.002652	Valid
0.016	OFF	6.89	1.65×10^3	0.000091–11.200642	Ejected at 0.272 yr

Here, $\text{RMS}_{0:T}$ is the time-averaged RMS separation from `ias15`, and $E_{\max}$ is the maximum fractional energy drift. “ON” and “OFF” refer to adaptive sub-stepping.

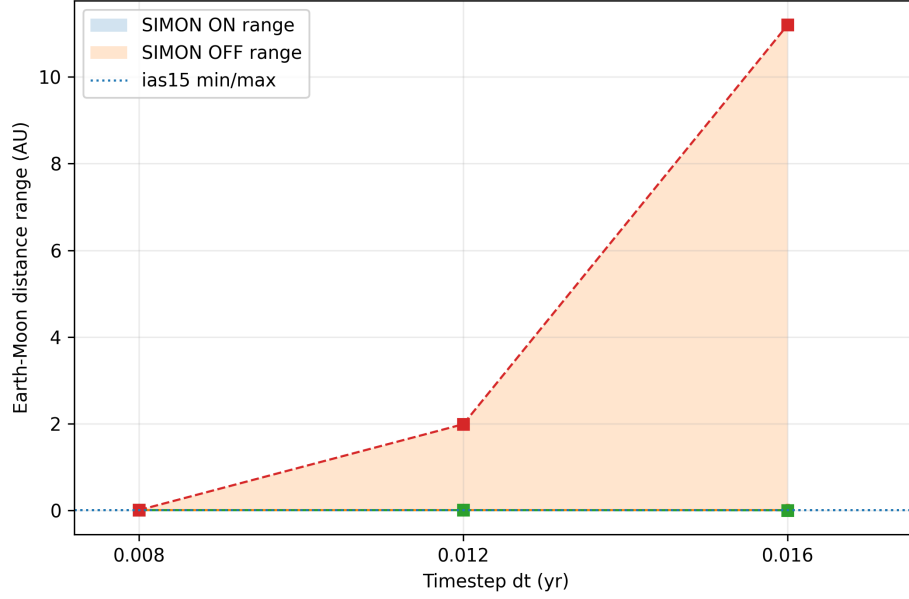


Figure 12: Earth–Moon distance range in the forward real-system validation.

The forward result supports the same conclusion as the synthetic ablation in Section 5.1. Adaptive sub-stepping is a stability mechanism: with it enabled, SIMON preserves the Earth–Moon bound state over 100 yr for all tested timesteps, with energy drift below $10^{-4}\%$. Without adaptive sub-stepping, the rollout first degrades, then loses the lunar orbit, and finally ejects at the timestep frontier.

A.2 Historical 1926–2026 Consistency Check

As a second check, we initialized the system from NASA/JPL Horizons on 1926-01-01 and integrated forward for approximately 100 yr. The Horizons reference samples span 1926-01-01 to 2025-12-19 because the 14-day sampling interval does not land exactly on 2026-01-01. We compare three trajectories: Horizons ephemeris samples, a simplified three-body *ias15* rollout from the same 1926 state, and SIMON at $dt = 0.016$ yr.

This comparison separates two effects. Horizons–*ias15* estimates the missing-physics gap caused by using a simplified Sun–Earth–Moon model, while SIMON–*ias15* measures SIMON’s error under the same simplified three-body equations.

Table 10: Historical Sun–Earth–Moon validation over approximately 100 yr.

Comparison	$\text{RMS}_{0:T}$ (AU)	Final RMS (AU)	Max RMS (AU)	Earth–Moon range (AU)	$E_{\max}$ (%)
Horizons– <i>ias15</i>	1.80×10^{-2}	3.12×10^{-2}	3.22×10^{-2}	0.002384–0.002718	7.53×10^{-6}
Horizons–SIMON	5.86×10^{-2}	4.42×10^{-2}	1.07×10^{-1}	0.002306–0.002742	8.15×10^{-5}
SIMON– <i>ias15</i>	4.45×10^{-2}	1.31×10^{-2}	8.27×10^{-2}	0.002306–0.002742	8.15×10^{-5}

The historical result is consistent with the forward $T = 100$ validation at the same

timestep. SIMON remains bounded, preserves the lunar distance scale (0.002306–0.002742 AU), and has $\text{RMS}_{0:T} = 4.45 \times 10^{-2}$ AU relative to `ias15`, matching the forward $dt = 0.016$ error scale. The comparison to Horizons should therefore be interpreted as an ephemeris-consistency check, not as a claim that a three-body model reproduces the full Solar System. The main physical result is that SIMON preserves the Sun–Earth and Earth–Moon structure over a historical century-scale rollout.

Appendix B: Force-Level NN Validation

To isolate the learned component of `SIMON`, we remove trajectory-level effects such as chaos, timestep drift, and phase error, and test the force law directly. Building on the softened-force formulation in Section 4.2, we sampled pair separations logarithmically across the close-encounter range and compared three force magnitudes: exact Newtonian gravity, the No-NN softened-force baseline, and the `SIMON` NN-corrected softened force. Relative force error is computed as $|F_{\text{model}} - F_{\text{Newt}}|/|F_{\text{Newt}}|$. This is not a trajectory rollout; it is a local force-level test of whether the learned scalar correction performs its intended role.

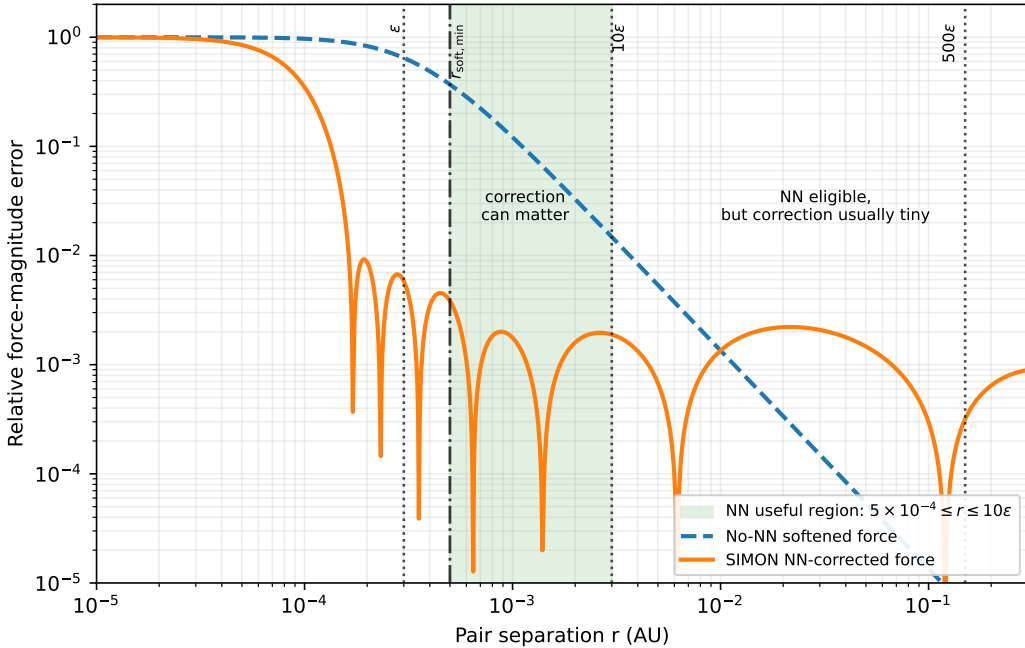


Figure 13: Relative force-magnitude error against exact Newtonian gravity.

Table 11: Force-error reduction for the representative mass pair.

Regime	Mean No-NN	Mean SIMON	Mean reduction	Max No-NN	Max SIMON
$r < 0.15$ AU	3.86×10^{-1}	2.17×10^{-1}	$1.78\times$	1.00	9.98×10^{-1}
$5 \times 10^{-4} \leq r < 0.15$ AU	3.92×10^{-2}	1.31×10^{-3}	$29.89\times$	3.69×10^{-1}	3.84×10^{-3}

Table 12: Mass robustness in the operational correction regime.

(m_i, m_j)	Mean No-NN	Mean SIMON	Mean reduction	Max No-NN	Max SIMON
(1.0, 0.01)	3.92×10^{-2}	1.31×10^{-3}	29.89×	3.69×10^{-1}	3.84×10^{-3}
(1.0, 0.5)	3.92×10^{-2}	9.07×10^{-4}	43.22×	3.69×10^{-1}	2.89×10^{-3}
(0.7, 0.4)	3.92×10^{-2}	8.82×10^{-4}	44.45×	3.69×10^{-1}	2.55×10^{-3}

The result confirms that the NN correction performs the local task for which it was trained. The two-regime structure in Table 11 reflects the design of SIMON’s safety gate: for $r < 5 \times 10^{-4}$ AU the integrator falls back to direct Newtonian force, bypassing the NN. The full close-eligible range ($r < 0.15$ AU, Row 1) therefore includes a mixed region where the NN correction is negligible (large r) and where the fallback fires (very small r), yielding a modest mean reduction of 1.78×. The operationally relevant comparison is Row 2: in the correction regime where the NN is both eligible and active ($5 \times 10^{-4} \leq r < 0.15$ AU), the mean relative force error falls from 3.92×10^{-2} for the No-NN softened baseline to 1.31×10^{-3} for SIMON, a 29.89× reduction. The same pattern holds across representative in-training mass pairs, supporting the interpretation in Section 5.1.4 that the NN correction is an accuracy refinement rather than the primary stability mechanism.

Appendix C: SIMON Architecture Schematic

Figure 14 summarises the full SIMON workflow. The schematic is intended as a compact visual map of the method rather than an additional result.

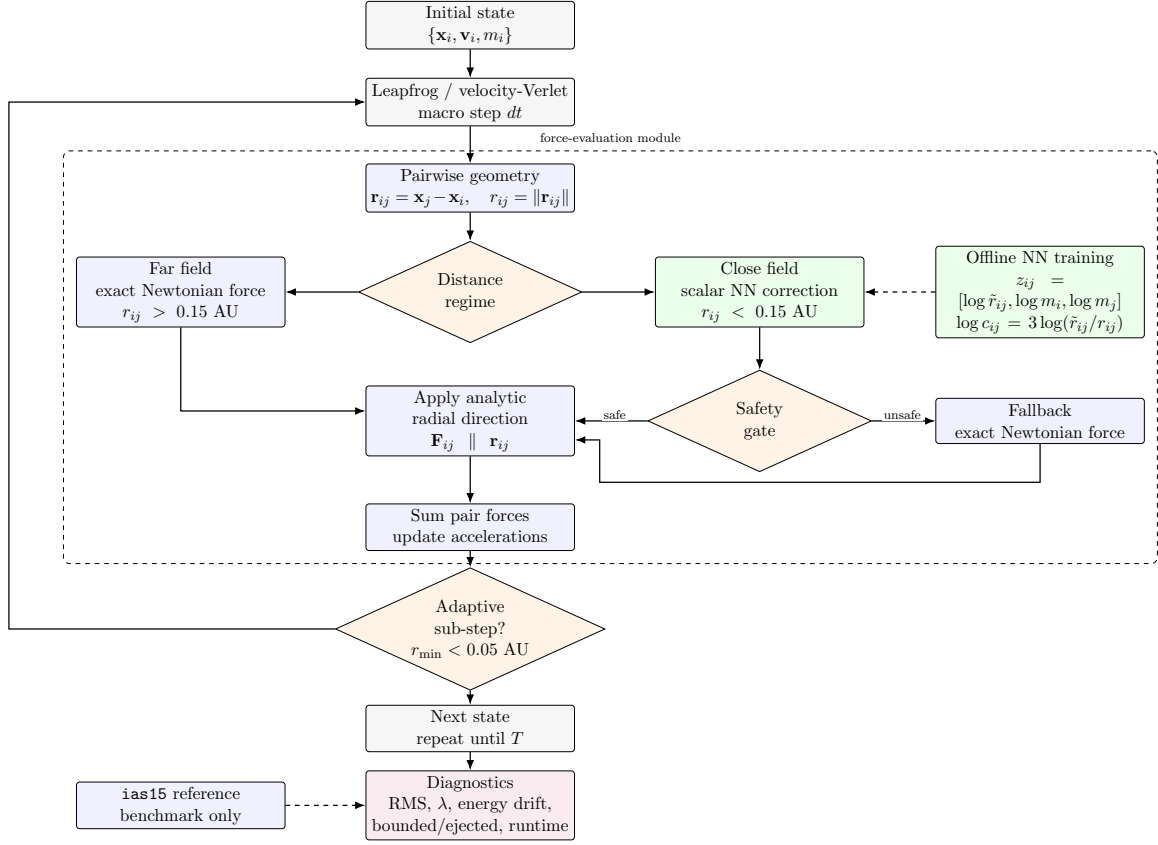


Figure 14: Architecture of SIMON.

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